

Melnick Nebart

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EDUCATION

Columbia University

New York, NY | 06.2027

Major: Astrophysics

Relevant Coursework: Observational Astronomy; Our Magnetic Universe: An Introduction to Astrophysical Magnetism; Quantum Mechanics I–II; Electromagnetic Waves & Optics; Electricity & Magnetism; Mechanics; Astrophysics I–II; Astronomy Laboratory; Introduction to Computing for Engineering and Applied Science; Differential Equations & Linear Algebra; Calculus I–IV.

Research Interests: Star formation and stellar populations; high-energy and AGN astrophysics; galactic dynamics; planetary surface processes and remote sensing; astronomical instrumentation; scientific computing and machine learning applications in astronomy.

RESEARCH EXPERIENCE

Lunar and Planetary Institute (LPI/USRA)

Houston, TX | 06.2026-09.2026

Research Intern [Supervisor: Dr Paul Schenk]

- Developed a Python-based surface-analysis workflow for Galileo- and Cassini-derived digital terrain models spanning approximately 9–500 m/pixel, combining ROI quality control, detrending, structure functions, Hurst exponent fitting, breakpoint detection, and power spectral density analysis.
- Compared scale-dependent topographic roughness across Europa, Ganymede, and Enceladus, analyzing multiple terrain classes including ridged plains, chaos, craters, ejecta, grooved terrain, fractured plains, and funicular terrain.
- Produced a research abstract, figures, and comparative results connecting Europa surface roughness to interpretation of surface clutter for NASA's Europa Clipper REASON radar instrument.

Space Telescope Science Institute (STScI)

Baltimore, MD | 06.2025-09.2025

Research Intern [Supervisor: Dr Rebecca Levy]

- Analyzed JWST/MIRI imaging of the starburst galaxy M82 using Python, Astropy, Photutils, SkyCoord, and DS9 to identify and characterize embedded young star-cluster candidates.
- Cross-matched MIRI-selected regions against HST/ACS, JWST/NIRCam, NICMOS, and submillimeter catalogs, constructing a multiwavelength census of obscured and previously detected stellar populations.
- Investigated MIRI-only sources as candidate highly embedded clusters that are difficult to detect at shorter wavelengths, extending the census of star formation in M82.
- Presented findings at the STScI Summer Research Symposium and contributed to a Research Notes of the AAS manuscript in preparation.

Columbia University: Research Skills

New York, USA | 01.2025-05.2025

Researcher [Supervisor: Alexandra Masegain, Dr David Hefland]

- Retrieved and cleaned Gaia DR3 data using Astroquery and Astropy, filtering astrometric, photometric, radial-velocity, and proper-motion measurements for candidate GD-1 stellar-stream members.
- Integrated stellar orbits over ± 2 Gyr with the GALA Python package to examine orbital convergence and reconstruct the stream's dynamical evolution within a Milky Way gravitational potential.
- Compared Plummer and NFW progenitor models against observed stream morphology to evaluate whether GD-1 originated from a disrupted star cluster or dwarf galaxy.

Columbia Physics Scholars Program

New York, USA | 01.2024-05.2024

Research Assistant [Supervisor: John Staunton]

- Processed and analyzed 500+ GB of Chandra X-ray Observatory data using CIAO, DS9, and Python to search for candidate X-ray counterparts associated with gravitational-wave sources.
- Investigated possible dual active galactic nuclei and supermassive black-hole merger systems through source detection, image filtering, counterpart identification, and observational-data validation.

Columbia Space Initiative

New York, USA | 09.2024-Present

Research Assistant -Microgravity Simulation (Engineering Focus)

- Designed and tested components of a custom three-axis clinostat intended to simulate microgravity environments for biological experiments.
- Created and refined mechanical components using SolidWorks and OpenSCAD while contributing to rotational-system integration, prototyping, and stability testing.
- Collaborated with a multidisciplinary student research team to translate experimental requirements into a functioning laboratory-scale microgravity simulator.

ASTRONOMY & INSTRUMENTATION PROJECTS

Fiber-Fed Multi-Object Spectrograph

New York, USA | 01.2026-05.2026

Columbia Observational Astronomy [Supervisor: Dr David Schminovich]

- Designed and constructed a multi-object fiber-fed spectrograph for Columbia's Celestron EdgeHD 14 telescope, using gnomonic projections to convert SIMBAD sky coordinates into millimeter-scale focal-plane positions.
- Fabricated custom aluminum plug plates and assembled nine 100- μ m optical fibers, integrating the fiber train with a bench spectrograph for simultaneous target acquisition.
- Calibrated the instrument with white-light and neon sources, successfully transmitting light through 5–7 fibers simultaneously and measuring a reciprocal linear dispersion of approximately 8.5 nm/mm.
- Diagnosed approximately 3–5 mm focal-plane alignment offsets during laboratory and on-sky testing, motivating higher-precision machining and improved ferrule alignment for future iterations.

Rutherford Dome Telescope Upgrades

New York, USA | 01.2026-05.2026

Columbia Observational Astronomy [Supervisor: Dr David Schminovich]

- Installed and integrated a PrimaLuce Lab ESATTO robotic focuser, ARCO field de-rotator, and ZWO imaging system on Columbia's Celestron EdgeHD 14 telescope.
- Troubleshooted communication among PLAY, ASCOM, Planewave, and telescope hardware, resolving mount connectivity, calibration, and incorrect de-rotation direction errors.
- Reduced star-trail length by 77.78% in 5-minute M82 exposures by enabling field de-rotation, demonstrating improved long-exposure imaging performance on the Alt-Az telescope.

COMPUTATIONAL PROJECTS

Infectious Disease Cellular-Automata Simulator

New York, USA | 01.2026-05.2026

ENGI E1006: Introduction to Computing for Engineering & Applied Science | Spring 2026

- Developed an object-oriented Python cellular-automata model simulating infectious-disease propagation across a 150×150 New York City grid.
- Implemented stochastic infection, recovery, and mortality using Cell and Map classes, nearest-neighbor interactions, discrete time steps, and a normally distributed mortality function.
- Visualized and quantified susceptible, infected, and resistant populations using NumPy and Matplotlib while varying virality, recovery time, and mortality parameters to compare outbreak behavior.

NYC Air Quality Data Explorer

New York, USA | 01.2026-05.2026

ENGI E1006 | Spring 2026

- Built a Python command-line application for querying NYC PM2.5 measurements by ZIP code, UHF region, borough, or date.
- Parsed and linked multiple CSV datasets using dictionary-based data structures mapping geographic IDs, ZIP codes, boroughs, dates, and PM2.5 observations.
- Implemented date normalization and multi-stage lookup logic to retrieve and format geographic air-quality records from user-defined search criteria.

PUBLICATIONS AND RESEARCH OUTPUTS

Nebart, M., Levy, R. C. Identifying and Cross-Matching Star Clusters in M82 with JWST/MIRI. Research Notes of the AAS manuscript in preparation.

Nebart, M., Schenk, P. Europa, Enceladus and Ganymede: Comparative Topography of Active Icy Worlds. Lunar and Planetary Institute Summer Intern Program research abstract, 2026.

Manyan, S., Nebart, M., Rodgers, N. Designing and Installing a Fiber-fed Spectrograph in the Pupin Rutherford Dome for Multi-Object Spectroscopy. Columbia University Observational Astronomy research report, 2026.

PRESENTATIONS

Nebart, M. (2026). "Europa, Enceladus and Ganymede: Comparative Topography of Active Icy Worlds". Oral presentation delivered at the Space, Lunar and Planetary Institute Summer Intern Program Research Symposium, Houston, TX, 2026.

Nebart, M. (2025). Identifying Embedded Star Clusters in the Starburst Galaxy M82 Using JWST/MIRI. Oral presentation delivered at the Space Telescope Science Institute (STScI) Summer Research Symposium, Baltimore, MD.

TEACHING EXPERIENCE

Department of Astronomy, Columbia University

New York, USA | 01.2025-05.2025

Teaching Assistant for ASTR UN1234: Universal Timekeeper[David Helfand]

- Graded and provided feedback on assignments for 60+ students, supporting consistent evaluation of quantitative and conceptual astronomy coursework.
- Supported student learning through course communication and explanation of astronomy concepts and assignments.

LEADERSHIP AND OUTREACH EXPERIENCES

Equity Leaders Program- Kenya

04.2023- Present

College Counselling Study Group Leader | June 2026- Present

College Counselling Consultant | April 2023- June 2026

- Lead and support college-counselling mentors serving students across East and Central Africa, monitoring mentor-mentee engagement and resolving application-related challenges.
- Mentored 10 students each year from Kenya, Uganda, Rwanda, and the Democratic Republic of the Congo through college selection, essays, financial aid, and application strategy.
- Coordinated mentor communication, progress reporting, and workshops to strengthen consistency and accountability across the counselling program.

Columbia Residential Life

New York, USA | 2025- Present

Residential Advisor

- Support an undergraduate residential community through advising, conflict mediation, emergency response, policy education, and referrals to university resources.
- Design and facilitate academic, wellness, and social programming to strengthen community engagement and resident belonging.
- Respond to student concerns through an on-call duty rotation while coordinating with Residential Life professional staff and campus resources.

Kenya Space Education Summit

Nairobi, Kenya | 07.2024-08.2024

Co-Coordinator

- Coordinated 10+ scientific presentations and workshops in space science, satellite technology, and STEM education for 200+ students and educators.
- Managed speaker communication, programming, logistics, and participant engagement in collaboration with Columbia Global Center Nairobi.
- Analyzed feedback from 150+ participants to evaluate learning outcomes and inform recommendations for future space-education programming.

Equity Global Scholars Board

New York, USA | 01.2024 - Present

First Year Representative | *USA Representative*

- Represented 100+ first-year Equity Global Scholars, facilitating engagement and mentorship initiatives to enhance community integration and support.
- Provided mentorship to 15 incoming scholars, aiding in their transition to college and academic success.

NAFSA NGO Youth Rep to the United Nations

New York, USA | 12/2024- Present

Social Media Coordinator

- Served as Social Media Coordinator for the NAFSA NGO Youth Reps at the United Nations, leading the development and execution of digital strategy to amplify global education and migration-related initiatives.
- Collaborated with UN event organizers and NAFSA representatives to create and share timely content for high-level NGO events at the UN, fostering public engagement and awareness.

ACTIVITIES AND ACADEMIC EXPERIENCES

Columbia Global Centre Nairobi

Nairobi, Kenya | 07.2024-09.2024

Programs Assistant Intern

- Coordinated logistics for 15+ events, including venue booking and catering, enhancing event efficiency and attendee satisfaction, including the Kenya Space Education Summit
- Conducted in-depth research on 5+ relevant topics, informing program development and enhancing community outreach efforts.
- Designed and distributed promotional materials, increasing program visibility and attendance by 20%.

Columbia University Housing

New York, USA | 01.2025-05.2025

Operations Assistant

- Managed and organized housing data in Asana, ensuring accurate record-keeping and efficient information access for the department.
- Collaborated with housing staff to streamline workflows, improving operational efficiency and enhancing resident satisfaction.
- Assisted in coordinating housing assignments, ensuring alignment with occupancy standards and student preferences.

Columbia Scholastic Press Association

New York, USA | 03.2024, 2025

CSPA Redshirt Assistant

- Managed audio-visual coordination for a convention of 500+ attendees, troubleshooting technical issues to ensure seamless presentations.
- Organized logistics for 10+ events, achieving a 95% satisfaction rate from participants.
- Oversaw setup, staffing, and breakdown of registration stations, ensuring efficient operation throughout the event.

Columbia University Disability Services

New York, USA | 09.2023-12.2023

Transcriptionist

- Facilitated comprehensive note-taking support for 20+ students with disabilities in the Intro to Mechanics class, ensuring accessibility and inclusivity in academic materials.
- Received commendations from faculty for responsiveness and exceptional communication, leading to improved support services.

Campus Reels

New York | 2023

Content Creator

- Created engaging short-form video content showcasing student life, academics, and campus culture at Columbia University for prospective students.
- Scripted, filmed, and edited original video content that aligned with CampusReel's brand and storytelling goals.
- Promoted authentic student perspectives to enhance college access and transparency for high school applicants from diverse backgrounds.

Equity Bank Limited

Ongata Rongai, Kenya | 05.2022-08.2022

Financial Operations Intern

- Processed financial transactions for 100+ customers per day while maintaining accuracy and compliance with branch procedures.
- Maintained transaction and inventory records while assisting customers with routine banking services.
- Supported high-volume branch operations through customer service, reconciliation, and administrative assistance.

HONORS AND AWARDS

Kluge Scholarship	2023, 2024, 2025, 2026
Columbia Dean's List	2024
Columbia Global Learner's Scholarship	2024
HERMAN FAMILY Summer Funding Award	2024
Space Telescope Science Institute Space Astronomy Summer Program	2025
Lunar and Planetary Institute Summer Intern Program in Planetary Science	2026

SKILLS

Programming: Python, NumPy, SciPy, Pandas, Matplotlib, Astropy, Astroquery, Photutils, gala, scikit-learn, OpenCV, Pillow, ruptures

Astronomy & Planetary Science: CIAO, DS9, FITS, Aladin, SIMBAD, ISIS, kalasiris, JWST/MIRI, Chandra, Gaia DR3, photometry, catalog cross-matching, CMD analysis, orbit integration, DTMs, stereo topography, shape-from-shading

Quantitative Methods: Structure functions, Hurst exponent estimation, PSD analysis, breakpoint/change-point detection, RMS roughness, numerical integration, stochastic simulation, statistical analysis, data visualization

Instrumentation & Engineering: Astronomical spectroscopy, optical fibers, telescope operation, field de-rotation, robotic focusing, instrument calibration, SolidWorks, OpenSCAD, CAD/DXF design

Scientific Communication: LaTeX/AAS_TEX, technical writing, scientific posters, oral research presentations

ADDITIONAL INFORMATION

Languages: English (Fluent), Swahili (Fluent), French (Elementary)

Workshops: Chandra/CIAO Workshop 2025, Columbia Undergraduate Mathematical Modeling Workshop (CUMMW)(2025)

Memberships: National Society of Black Physicists(2024- Present), Society of Physics Students(2024-present)

Interests: African Traditional Music, Jazz, Drumming, Science fiction, Memes, cooking international cuisine, visiting museums, basketball